

BRIEFING ON SPECTRAL GEOMETRY

ABSTRACT.

1. INTRODUCTION

1.1. **“Can one hear the shape of a drum?”** In 1966, mathematician Mark Kac posed a now-famous question: “Can one hear the shape of a drum?” More formally, he asked whether the shape of a plane domain can be determined solely from the eigenvalues of its Laplacian operator with Dirichlet boundary conditions. To explore this, picture a drum with its head modeled as a bounded domain $\Omega \subset \mathbb{R}^2$ with a smooth boundary. When struck, the drum produces a sound defined by its natural frequencies of vibration. These frequencies are the eigenvalues of the Laplacian operator Δ on Ω , solutions to the equation

$$-\Delta u = \lambda u \quad \text{in } \Omega,$$

where u satisfies $u = 0$ on $\partial\Omega$, representing a fixed edge. The spectrum of Ω is the sequence $\{\lambda_k\}_{k=1}^{\infty}$ of these eigenvalues, ordered increasingly, encoding the drum’s vibrational symphony.

Kac’s question translates to: If two domains Ω_1 and Ω_2 share the same spectrum, are they isometric—identical up to rigid motions? For decades, this puzzled mathematicians, with hints that properties like area or perimeter might be spectral invariants. Yet, in 1992, Gordon, Webb, and Wolpert overturned this hope by constructing two planar domains that are isospectral but not isometric. Thus, the answer is no: one cannot always hear the shape

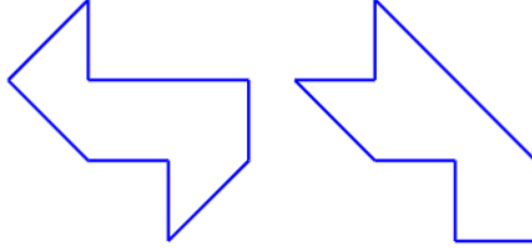


FIGURE 1. Isospectral plane domains and surfaces via Riemannian orbifolds
— *Inventiones mathematicae*

of a drum. This counterintuitive result—where distinct shapes resonate identically—reveals the intricate dance between geometry and analysis.

Spectral geometry, however, extends far beyond this riddle. It probes how a space's geometric traits, such as volume, curvature, or dimension, influence the spectra of operators like the Laplacian, and what these spectra, in turn, reveal. While the drum's exact shape eludes us, its spectrum still whispers secrets of its nature, inviting deeper exploration into this harmonious interplay of form and sound.

1.2. Weak Dirichlet Spectral Problem. Let $\Omega \subset \mathbb{R}^n$ be a bounded domain with a smooth boundary $\partial\Omega$. The weak Dirichlet spectral problem for the Laplace operator seeks a function u and a real number λ that satisfy specific conditions. Specifically, we aim to: Find $u \in H_0^1(\Omega)$ and $\lambda \in \mathbb{R}$ such that

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = \lambda \int_{\Omega} uv \, dx \quad \text{for all } v \in H_0^1(\Omega).$$

Here, $H_0^1(\Omega)$ is the Sobolev space that vanishes on the boundary $\partial\Omega$ (in the sense of traces). This weak formulation arises from the strong form $-\Delta u = \lambda u$ with the Dirichlet

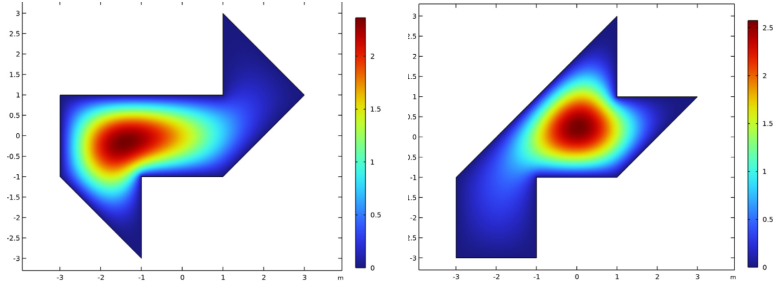


FIGURE 2. Dirichlet Problem

boundary condition $u = 0$ on $\partial\Omega$, where integration by parts eliminates boundary terms due to the zero trace of test functions v .

1.3. Weak Neumann Spectral Problem. Similarly, for the weak Neumann spectral problem, we consider the same domain $\Omega \subset \mathbb{R}^n$ with a smooth boundary $\partial\Omega$. The problem is defined as follows:

Find $u \in H^1(\Omega)$ and $\lambda \in \mathbb{R}$ such that

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = \lambda \int_{\Omega} uv \, dx \quad \text{for all } v \in H^1(\Omega).$$

In this case, $H^1(\Omega)$ is the Sobolev space without any imposed boundary conditions on u . This formulation corresponds to the strong form $-\Delta u = \lambda u$ with the Neumann boundary condition $\frac{\partial u}{\partial n} = 0$ on $\partial\Omega$.

The distinction lies in the choice of Sobolev space. For the Dirichlet problem, u and the test functions v belong to $H_0^1(\Omega)$, enforcing the essential boundary condition $u = 0$ on $\partial\Omega$. For the Neumann problem, u and v are in $H^1(\Omega)$, and the boundary condition is naturally satisfied within the weak formulation.

1.4. Dirichlet Laplacian. To formalize, consider the operator $-\Delta_D : D(-\Delta_D) \rightarrow L^2(\Omega)$. Its domain $D(-\Delta_D)$ consists of $u \in H_0^1(\Omega)$ such that there exists $f \in L^2(\Omega)$ with

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx \quad \text{for all } v \in H_0^1(\Omega),$$

and $-\Delta_D u = f$. Self-adjointness follows from the symmetry:

$$(-\Delta_D u, v) = \int_{\Omega} f v \, dx = \int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} \nabla v \cdot \nabla u \, dx = (u, -\Delta_D v),$$

for $u, v \in D(-\Delta_D)$, using integration by parts and the boundary condition $u = 0$ on $\partial\Omega$, which eliminates boundary terms.

1.5. Neumann Laplacian. The domain for Neumann Laplacian $-\Delta_N : D(-\Delta_N) \rightarrow L^2(\Omega)$ is defined as

$$D(-\Delta_N) = \left\{ u \in H^1(\Omega) \mid \exists f \in L^2(\Omega) \text{ such that } \int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx \text{ for all } v \in H^1(\Omega) \right\}.$$

Below is a description of the Dirichlet and Neumann Laplacians using the variational principle for a quadratic form and the Rayleigh quotient, tailored to the user's query. The explanation is complete, self-contained, and leverages the key insights from the thinking trace, while omitting extraneous details.

1.6. Rayleigh Quotient. Both the Dirichlet and Neumann Laplacians are linked to the same **quadratic form**:

$$Q(u) = \int_{\Omega} |\nabla u|^2 dx$$

This form represents the "energy" associated with a function u , measuring the square of its gradient over the domain Ω . The Rayleigh quotient, which is central to the variational characterization, is defined as:

$$R(u) = \frac{Q(u)}{\|u\|_{L^2}^2} = \frac{\int_{\Omega} |\nabla u|^2 dx}{\int_{\Omega} u^2 dx}, \quad u \neq 0$$

The eigenvalues of these operators correspond to the critical values of $R(u)$, and the eigenfunctions are the functions at which these critical values are attained, subject to appropriate constraints.

1.7. Compactness of $(-\Delta_D - 1)^{-1}$. $(-\Delta - 1)^{-1}$ is bounded from $L^2(\Omega)$ to $H_0^1(\Omega)$. An operator is compact if it maps bounded sets into relatively compact sets. Since $(-\Delta_N - 1)^{-1} : L^2(\Omega) \rightarrow H_0^1(\Omega)$, and we are interested in its action within $L^2(\Omega)$, we examine the embedding:

$$H_0^1(\Omega) \hookrightarrow L^2(\Omega).$$

The Rellich-Kondrachov theorem states that for a bounded domain $\Omega \subset \mathbb{R}^n$ with smooth boundary, this embedding is **compact**.

1.8. Discrete Spectrum Properties. Since $-\Delta_D$ is self-adjoint and its resolvent $(-\Delta_D + 1)^{-1}$ is compact, we apply the spectral theorem for compact self-adjoint operators. The operator $(-\Delta_D + 1)^{-1}$ has a spectrum consisting of:

- (1) A sequence of real eigenvalues $\{\mu_k\}$, with $\mu_k \rightarrow 0$ as $k \rightarrow \infty$,
- (2) Each $\mu_k \neq 0$ having finite multiplicity,
- (3) The only accumulation point of the spectrum being 0.

Because $-\Delta_D + 1$ is positive definite (its eigenvalues are $\lambda_k + 1 > 1$), the eigenvalues μ_k are positive. The eigenvalues of $-\Delta_D$ are then:

$$\lambda_k = \frac{1}{\mu_k} - 1.$$

Since $\mu_k > 0$ and $\mu_k \rightarrow 0$ as $k \rightarrow \infty$, it follows that $\lambda_k > 0$ and $\lambda_k \rightarrow \infty$. Moreover, the finite multiplicity of each μ_k implies that each λ_k has finite multiplicity.

The Dirichlet Laplacian $-\Delta_D$ on a bounded domain $\Omega \subset \mathbb{R}^n$ with a smooth boundary has a **discrete spectrum**. This spectrum consists of a sequence of positive eigenvalues:

$$0 < \lambda_1 \leq \lambda_2 \leq \lambda_3 \leq \cdots \rightarrow \infty,$$

where:

- (1) Each λ_k is isolated and has finite multiplicity,
- (2) The eigenvalues accumulate only at infinity.
- (3) The eigenfunctions form orthonormal basis of $L^2(\Omega)$.

1.9. Proving $\lambda_1 > 0$. The first eigenvalue λ_1 can be characterized using the **Rayleigh quotient**. This is a powerful tool in spectral theory, and it states that:

$$\lambda_1 = \min_{u \in H_0^1(\Omega), u \neq 0} \frac{\int_{\Omega} |\nabla u|^2 dx}{\int_{\Omega} u^2 dx}.$$

The **Poincaré inequality** is the key here. For a bounded domain Ω , it asserts that there exists a constant $C > 0$, depending on Ω , such that for all $u \in H_0^1(\Omega)$:

$$\int_{\Omega} u^2 dx \leq C \int_{\Omega} |\nabla u|^2 dx.$$

Since $C > 0$, it follows that:

$$\frac{\int_{\Omega} |\nabla u|^2 dx}{\int_{\Omega} u^2 dx} \geq \frac{1}{C} > 0.$$

This inequality holds for all $u \in H_0^1(\Omega)$, $u \neq 0$. Since λ_1 is the minimum value of the Rayleigh quotient over all such u , we have:

$$\lambda_1 = \min_{u \in H_0^1(\Omega), u \neq 0} \frac{\int_{\Omega} |\nabla u|^2 dx}{\int_{\Omega} u^2 dx} \geq \frac{1}{C}.$$

1.10. Min-max Principle. Higher eigenvalues λ_k (for $k = 2, 3, \dots$) can be characterized using:

$$(1.1) \quad \lambda_k = \min_{\substack{V \subset H_0^1(\Omega) \\ \dim V = k}} \max_{u \in V, u \neq 0} R(u)$$

, where

$$R(u) = \frac{\int_{\Omega} |\nabla u|^2 dx}{\int_{\Omega} u^2 dx}.$$

Here, the minimum is taken over all k -dimensional subspaces V of $H_0^1(\Omega)$, and the maximum is over non-zero functions in V . This formulation systematically generates the discrete spectrum of $-\Delta_D$, with $\lambda_k \rightarrow \infty$ as $k \rightarrow \infty$.

1.11. Spectral of The Neumann Laplacian. The eigenvalues are $0 = \mu_1 < \mu_2 \leq \mu_3 \leq \dots$, where $\mu_1 = 0$ is given by constant functions and $\mu_2 > 0$ is proved by Poincaré inequality for function with zero mean value. We also have

$$(1.2) \quad \mu_k = \min_{\substack{V \subset H^1(\Omega) \\ \dim V = k}} \max_{u \in V, u \neq 0} R(u)$$

, where

$$R(u) = \frac{\int_{\Omega} |\nabla u|^2 dx}{\int_{\Omega} u^2 dx}.$$

We need to consider spectral problems with mixed Dirichlet-Neumann boundary conditions, known as Zaremba problems. Let Ω be a bounded domain in $\mathbb{R}^d$ with a Lipschitz boundary $\partial\Omega$, decomposed into Dirichlet boundary $\Gamma := \partial_D\Omega$ and Neumann boundary $\partial_N\Omega := \partial\Omega \setminus \Gamma$. We assume that both boundaries consist of a finite number of connected components and that the interface is sufficiently regular for $d \geq 3$. The mixed Dirichlet-Neumann spectral problem is given by:

$$\begin{cases} -\Delta u = \lambda u & \text{in } \Omega, \\ u = 0 & \text{on } \partial_D\Omega, \\ \partial_n u = 0 & \text{on } \partial_N\Omega. \end{cases}$$

1.12. Zaremba problem. We define the space

$$C_{0,\Gamma}^\infty(\Omega) := \{u \in C^\infty(\Omega) : \text{supp } u \cap \bar{\Gamma} = \emptyset\}$$

and the Sobolev space $H_{0,\Gamma}^1(\Omega)$ as its completion in the $H^1(\Omega)$ norm. The Zaremba Laplacian $-\Delta_\Gamma^Z$ is defined via the Friedrichs extension with the domain

$$\text{Dom}(-\Delta_\Gamma^Z) = \{u \in H_{0,\Gamma}^1(\Omega) : \Delta u \in L^2(\Omega) \text{ and } \partial_n u \sim 0 \text{ on } \partial_N\Omega\}.$$

The bilinear form $Q^Z[u, \nu]$ for the mixed problem aligns with those for Dirichlet and Neumann Laplacians, differing only in its domain $H_{0,\Gamma}^1(\Omega)$. The eigenvalues $\lambda_k^Z(\Omega, \Gamma)$ are determined by the variational principle:

$$(1.3) \quad \lambda_k^Z(\Omega, \Gamma) = \min_{\substack{\mathcal{L} \subset H_{0,\Gamma}^1(\Omega) \\ \dim \mathcal{L} = k}} \max_{u \in \mathcal{L} \setminus \{0\}} \frac{\|\nabla u\|_{L^2(\Omega)}^2}{\|u\|_{L^2(\Omega)}^2}, \quad k \in \mathbb{N}.$$

1.13. Counting Function. Let A be a self-adjoint and semi-bounded from below operator with a discrete spectrum consisting of eigenvalues $\lambda_1 \leq \lambda_2 \leq \dots$. The eigenvalue counting function of A is the function

$$\mathcal{N} : \mathbb{R} \rightarrow \mathbb{N}_0$$

defined by

$$\mathcal{N}(\lambda) = \mathcal{N}^A(\lambda) := \#\{j : \lambda_j(A) \leq \lambda\}.$$

1.14. Glazman's Lemma. Consider the weak spectral problem associated with a symmetric bilinear form $\mathcal{Q}$ that is semi-bounded from below. The counting function $M^{\mathcal{Q}}(\lambda)$ for the corresponding weak eigenvalues satisfies

$$M^{\mathcal{Q}}(\lambda) = \max_{\substack{\mathcal{L} \subset \text{Dom}(\mathcal{Q}) \\ R[u] \leq \lambda \forall u \in \mathcal{L} \setminus \{0\}}} \dim \mathcal{L},$$

where $\mathcal{L}$ is a finite-dimensional linear subspace of $\text{Dom}(\mathcal{Q})$, and $R[u]$ denotes the Rayleigh quotient.

2. WEYL'S LAW

2.1. Domain Monotonicity.

Theorem 2.1 (Domain Monotonicity for the Dirichlet Laplacian). *Let $\Omega_1 \subset \Omega_2$ be two bounded domains in $\mathbb{R}^d$. Then, their Dirichlet eigenvalues satisfy*

$$\lambda_k(\Omega_2) \leq \lambda_k(\Omega_1) \quad \text{for all } k \in \mathbb{N}.$$

Proof. Consider the natural embedding $H_0^1(\Omega_1) \hookrightarrow H_0^1(\Omega_2)$. For any function $u \in H_0^1(\Omega_1)$, define its extension $\tilde{u} \in H_0^1(\Omega_2)$ by setting $\tilde{u} = u$ in Ω_1 and $\tilde{u} = 0$ in $\Omega_2 \setminus \Omega_1$. It follows that the Rayleigh quotient remains invariant under this extension, i.e., $R_{\Omega_1}[u] = R_{\Omega_2}[\tilde{u}]$, where

$$R_{\Omega}[u] = \frac{\int_{\Omega} |\nabla u|^2 dx}{\int_{\Omega} u^2 dx}.$$

The inequality $\lambda_k(\Omega_2) \leq \lambda_k(\Omega_1)$ then follows directly from the variational characterization of Dirichlet eigenvalues, as given by equations (1.1) and (1.2), since the test function space for Ω_2 includes all functions from Ω_1 extended by zero. $\square$

Proposition 2.1 (Dirichlet-Neumann Bracketing, Version I). *Let $\Omega \subset \mathbb{R}^d$ be a bounded domain with a Lipschitz boundary, and let $\Gamma_1 \subset \Gamma_2$ be subsets of $\partial\Omega$. Then, the mixed*

Dirichlet-Neumann eigenvalues satisfy

$$\lambda_k^Z(\Omega, \Gamma_1) \leq \lambda_k^Z(\Omega, \Gamma_2) \quad \text{for all } k \in \mathbb{N},$$

where Dirichlet conditions are imposed on Γ_i and Neumann conditions on $\partial\Omega \setminus \Gamma_i$.

Proof. The result follows from the variational principle for mixed eigenvalue problems. The inclusion $H_0^1(\Omega, \Gamma_1) \hookrightarrow H_0^1(\Omega, \Gamma_2)$ holds, since functions in $H_0^1(\Omega, \Gamma_1)$, which vanish on Γ_1 , also vanish on the smaller set $\Gamma_1 \subset \Gamma_2$ and thus belong to $H_0^1(\Omega, \Gamma_2)$. As the Dirichlet boundary condition is imposed over a larger set in Γ_2 , the constraint on the test functions increases, leading to a larger eigenvalue. $\square$

2.2. Dirichlet-Neumann Bracketing with Internal Hypersurface. Now, consider a bounded domain $\Omega \subset \mathbb{R}^d$ equipped with the Dirichlet Laplacian $-\Delta_D$. Let $\Gamma \subset \Omega$ be a Lipschitz hypersurface, and define $\Omega' = \Omega \setminus \Gamma$, so that $\partial\Omega' = \partial\Omega \cup \Gamma$.

Proposition 2.2 (Dirichlet-Neumann Bracketing, Version II). *For the geometry described above, the eigenvalues satisfy*

$$\lambda_k^Z(\Omega', \partial\Omega) \leq \lambda_k(\Omega, \Gamma) \leq \lambda_k(\Omega') \quad \text{for all } k \in \mathbb{N}.$$

Proof. The first inequality, $\lambda_k(\Omega, \Gamma) \leq \lambda_k(\Omega')$, arises because imposing additional Dirichlet conditions on Γ restricts the test function space, increasing the eigenvalues. The second inequality, $\lambda_k^Z(\Omega', \partial\Omega) \leq \lambda_k(\Omega, \Gamma)$, follows from the fact that replacing Dirichlet conditions on Γ with Neumann conditions enlarges the test function space, reducing the eigenvalues. These bounds are consistent with the variational principle applied to the respective boundary conditions. $\square$

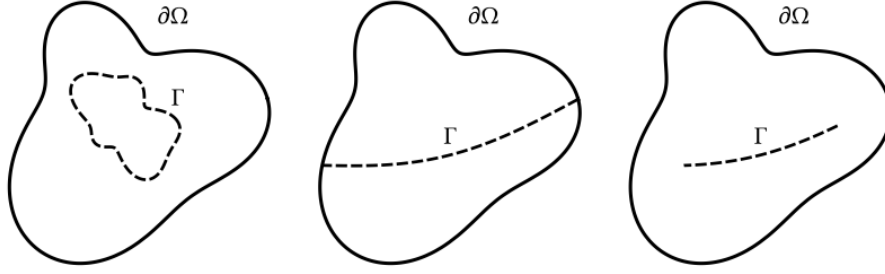


FIGURE 3. Three possible configurations of a hypersurface Γ within Ω . Left: Γ is a closed hypersurface. Middle: $\partial\Gamma = \partial\Omega$. Right: $\partial\Gamma \subset \Omega$.

2.3. Weyl's Law.

Theorem 2.2 (Weyl's Law). *Let $\Omega \subset \mathbb{R}^n$ be a bounded domain with smooth boundary, and let $-\Delta$ be the Dirichlet Laplacian on Ω . Denote the eigenvalues of $-\Delta$ by $0 < \lambda_1 \leq \lambda_2 \leq \lambda_3 \leq \dots \rightarrow \infty$, and define the eigenvalue counting function as $N(\lambda) = \#\{k : \lambda_k \leq \lambda\}$. Then, as $\lambda \rightarrow \infty$,*

$$N(\lambda) \sim \frac{\omega_n}{(2\pi)^n} \cdot \text{Vol}(\Omega) \cdot \lambda^{n/2},$$

where $\omega_n = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2}+1)}$ is the volume of the unit ball in $\mathbb{R}^n$.

2.4. Example: Rectangular Domain Satisfying Weyl's Law. Consider the rectangular domain $\Omega = [0, a] \times [0, b] \subset \mathbb{R}^2$ with Dirichlet boundary conditions. The eigenvalues of the Laplacian $-\Delta$ on Ω are explicitly given by:

$$\lambda_{m,n} = \pi^2 \left(\frac{m^2}{a^2} + \frac{n^2}{b^2} \right), \quad m, n \in \mathbb{N}.$$

The eigenvalue counting function is:

$$N(\lambda) = \# \left\{ (m, n) \in \mathbb{N}^2 : \pi^2 \left(\frac{m^2}{a^2} + \frac{n^2}{b^2} \right) \leq \lambda \right\}.$$

Weyl's Law predicts:

$$N(\lambda) \sim \frac{\omega_2}{(2\pi)^2} \cdot \text{Vol}(\Omega) \cdot \lambda \quad \text{as } \lambda \rightarrow \infty,$$

where $\omega_2 = \pi$ (volume of the unit ball in $\mathbb{R}^2$) and $\text{Vol}(\Omega) = ab$. Substituting these values:

$$N(\lambda) \sim \frac{\pi}{(2\pi)^2} \cdot ab \cdot \lambda = \frac{ab}{4\pi} \lambda.$$

The inequality $\frac{m^2}{a^2} + \frac{n^2}{b^2} \leq \frac{\lambda}{\pi^2}$ describes an ellipse in the (m, n) -plane. For large λ , the number of lattice points (m, n) inside this ellipse approximates its area. Rescaling variables $x = \frac{m}{a}$, $y = \frac{n}{b}$, the ellipse becomes a quarter-circle of radius $\sqrt{\lambda}/\pi$. The area in the (x, y) -plane is:

$$\text{Area} = \frac{1}{4} \pi \left(\frac{\sqrt{\lambda}}{\pi} \right)^2 = \frac{\lambda}{4\pi}.$$

Scaling back to the original (m, n) -plane (area scaled by ab):

$$N(\lambda) \sim ab \cdot \frac{\lambda}{4\pi} = \frac{ab}{4\pi} \lambda,$$

which matches Weyl's Law.

The rectangular domain Ω satisfies Weyl's Law, confirming that the asymptotic growth of $N(\lambda)$ depends only on the volume ab and the dimension $n = 2$.

Example: Rectangular Domain Satisfying Neumann Weyl's Law. Consider the rectangular domain $\Omega = [0, a] \times [0, b] \subset \mathbb{R}^2$ with **Neumann boundary conditions**. The eigenvalues of the Laplacian $-\Delta$ on Ω are given by:

$$\lambda_{m,n} = \pi^2 \left(\frac{m^2}{a^2} + \frac{n^2}{b^2} \right), \quad m, n \in \mathbb{N}_0 = \{0, 1, 2, \dots\},$$

where m and n are non-negative integers (excluding the trivial case $m = n = 0$ if the constant eigenfunction is disregarded). The eigenvalue counting function is:

$$N(\lambda) = \# \left\{ (m, n) \in \mathbb{N}_0^2 : \pi^2 \left(\frac{m^2}{a^2} + \frac{n^2}{b^2} \right) \leq \lambda \right\}.$$

2.5. Counting Function for Disconnected Regions. Let Ω_1 and Ω_2 be two disjoint (non-connected) bounded domains in $\mathbb{R}^n$ with smooth boundaries. Consider the Dirichlet Laplacian $-\Delta$ on their union $\Omega = \Omega_1 \sqcup \Omega_2$. The eigenvalues of $-\Delta$ on Ω are the union of the eigenvalues of $-\Delta$ on Ω_1 and Ω_2 , counted with multiplicity. The total eigenvalue counting function $N_\Omega(\lambda)$ for the disjoint union satisfies:

$$N_\Omega(\lambda) = N_{\Omega_1}(\lambda) + N_{\Omega_2}(\lambda),$$

where:

$$N_{\Omega_i}(\lambda) = \#\{k : \lambda_k(\Omega_i) \leq \lambda\}, \quad i = 1, 2.$$

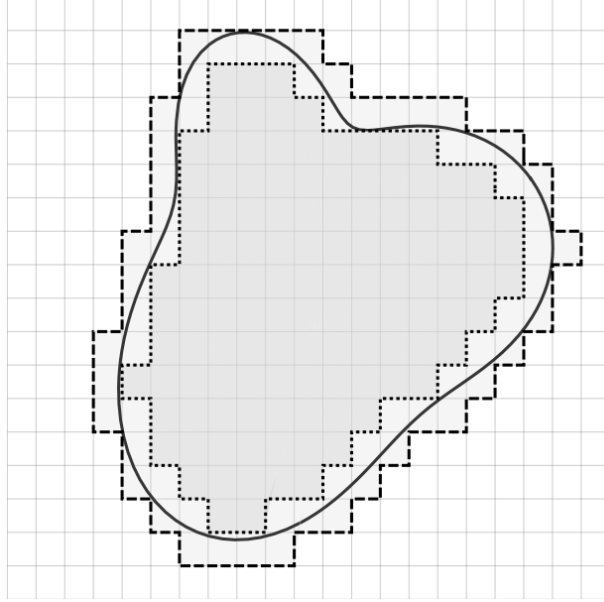


FIGURE 4. An example of a domain Ω , with its boundary shown as a solid curve, and corresponding domains $\Omega_{I,a}$ (darker shading, boundary is shown as a dotted line) and $\Omega_{E,a}$ (lighter shading, boundary is shown as a dashed line.)

2.6. Proof Sketch of Weyl's Law (Dirichlet Case).

Proof. Step 1: Cubes with Dirichlet/Neumann boundary conditions. Weyl's law is verified for cubes. The key difference lies in counting lattice points:

- *Dirichlet case:* Eigenvalues correspond to $\lambda_{m_1, \dots, m_d} = \pi^2 \sum_{j=1}^d \frac{m_j^2}{a_j^2}$ with $m_j \in \mathbb{N}$.
- *Neumann case:* $m_j \in \mathbb{N}_0$ (non-negative integers).

The counting function asymptotics match the volume term $\frac{\omega_d}{(2\pi)^d} \text{Vol}(\text{cube}) \lambda^{d/2}$.

Step 2: Almost disjoint unions of cubes. Let Ω be a domain represented as an *almost disjoint union* of cubes $\mathcal{K}$, i.e., $\Omega = \text{int}(\overline{\mathcal{K}})$ with $\partial\Omega \subset \partial\mathcal{K}$. Impose Dirichlet/Neumann conditions on $\tilde{\Omega} = \Omega \setminus \partial\mathcal{K}$. By Dirichlet-Neumann bracketing:

$$\mathcal{N}_{\tilde{\Omega}}^D(\lambda) \leq \mathcal{N}_{\Omega}^D(\lambda) \leq \mathcal{N}_{\tilde{\Omega}}^N(\lambda).$$

The result follows by summing counting functions over cubes and applying Step 1.

Step 3: Arbitrary bounded domains. For any bounded Ω , construct inner/outer approximations:

$$\Omega_{I,a} \subset \Omega \subset \Omega_{E,a},$$

where $\Omega_{I,a}, \Omega_{E,a}$ are almost disjoint unions of cubes with side a . By domain monotonicity:

$$\mathcal{N}_{\Omega_{I,a}}^D(\lambda) \leq \mathcal{N}_{\Omega}^D(\lambda) \leq \mathcal{N}_{\Omega_{E,a}}^D(\lambda).$$

Taking $\lambda \rightarrow \infty$ and then $a \rightarrow 0$:

$$\limsup_{\lambda \rightarrow \infty} \frac{\mathcal{N}_{\Omega}^D(\lambda)}{\lambda^{d/2}} \leq C_d \text{Vol}_d(\Omega_{E,a}), \quad \liminf_{\lambda \rightarrow \infty} \frac{\mathcal{N}_{\Omega}^D(\lambda)}{\lambda^{d/2}} \geq C_d \text{Vol}_d(\Omega_{I,a}),$$

with $\lim_{a \rightarrow 0} \text{Vol}_d(\Omega_{I,a}) = \lim_{a \rightarrow 0} \text{Vol}_d(\Omega_{E,a}) = \text{Vol}_d(\Omega)$. This completes the proof. $\square$

3. NODAL GEOMETRY OF EIGENFUNCTIONS

3.1. Courant's Nodal Domain Theorem. First established by Richard Courant in 1923 [1], this fundamental theorem connects spectral theory with geometric properties of eigenfunctions.

Definition 3.1 (Nodal Domain). *For an eigenfunction u_n of a differential operator $\mathcal{L}$, a **nodal domain** is a maximally connected subdomain $\Omega_i \subset \Omega$ where:*

$$u_n(x) > 0 \quad \forall x \in \Omega_i \quad \text{or} \quad u_n(x) < 0 \quad \forall x \in \Omega_i$$

The nodal set is defined as

$$\mathcal{N}(u_n) = \{x \in \Omega \mid u_n(x) = 0\}$$

3.2. Restriction of an Eigenfunction to a Nodal Domain. A non-trivial technical step in the proof of Courant's theorem is the following key result:

Theorem 3.1 (Restriction Property). *Let $u \in H_0^1(\Omega) \cap C(\Omega)$, and let $\Omega_1 \subset \Omega$ be a nodal domain of u . Then, $u|_{\Omega_1} \in H_0^1(\Omega_1)$.*

Theorem 3.2 (Courant's Nodal Domain Theorem). *Let $\Omega \subset \mathbb{R}^n$ be a bounded domain with smooth boundary. The k^{th} eigenfunction u_k of the Dirichlet Laplacian $-\Delta_D$ has at most k nodal domains.*

Proof. For eigenfunction u with $\lambda = \lambda(\Omega)$ and $(k+1)$ nodal domains $\{\Omega_i\}_1^{k+1}$, define test functions:

$$\psi_i(x) = \begin{cases} u(x) & x \in \Omega_i \\ 0 & \text{otherwise} \end{cases}$$

By Theorem 3.1, $\psi_i \in H_0^1(\Omega_i)$. For $\tilde{\Omega} = \bigcup_1^k \Omega_i$ and $\mathcal{L} = \text{Span}\{\psi_1, \dots, \psi_k\}$, any $\psi = \sum c_i \psi_i \in \mathcal{L}$ satisfies:

$$\|\nabla \psi\|_{L^2(\tilde{\Omega})}^2 = \lambda \|\psi\|_{L^2(\tilde{\Omega})}^2$$

Thus $\mathcal{R}_{\tilde{\Omega}}[\psi] = \lambda$ for all $\psi \in \mathcal{L} \setminus \{0\}$, implying $\lambda \geq \lambda_k(\tilde{\Omega})$ by variational principle. Domain monotonicity gives $\lambda_k(\tilde{\Omega}) > \lambda_k(\Omega)$ since $\Omega \setminus \tilde{\Omega} \supset \Omega_{k+1} \neq \emptyset$. Hence:

$$\lambda \geq \lambda_k(\tilde{\Omega}) > \lambda_k(\Omega).$$

□

Interesting version of proof that avoid using Theorem 3.1: The Courant Nodal Domain Theorem .

3.3. Nodal set density estimate.

Theorem 3.3 (Nodal set density estimate). *Let f be a non-trivial solution to $-\Delta f = \lambda f$ with $\lambda > 0$ in a domain $\Omega \subset \mathbb{R}^d$. Then the nodal set $\mathcal{N}(f)$ satisfies the $c_d\sqrt{\lambda}$ -density property, where $c_d = j_{d/2-1,1}^{-1}$ and $j_{m,1}$ denotes the first positive zero of the Bessel function J_m .*

Proof of Theorem 3.3. Let $\Omega' \Subset \Omega$ be a smooth bounded subdomain where f maintains constant sign. Without loss of generality, assume $f > 0$ in Ω' . Let $u_1 > 0$ be the principal Dirichlet eigenfunction in Ω' with eigenvalue $\lambda_1^D(\Omega')$.

Applying Green's formula to u_1 and f :

$$\begin{aligned} \langle (\lambda_1^D(\Omega') - \lambda)u_1, f \rangle_{L^2(\Omega')} &= \langle -\Delta u_1, f \rangle - \langle u_1, -\Delta f \rangle \\ &= - \int_{\partial\Omega'} (\partial_n u_1) f \, d\sigma + \int_{\partial\Omega'} u_1 (\partial_n f) \, d\sigma \\ &= - \int_{\partial\Omega'} (\partial_n u_1) f \, d\sigma \geq 0. \end{aligned}$$

Since $u_1|_{\partial\Omega'} = 0$ with $\partial_n u_1 \leq 0$ (by Hopf's lemma), and $f|_{\partial\Omega'} \geq 0$ by continuity, the boundary integral is non-negative. Given $\langle u_1, f \rangle_{L^2(\Omega')} > 0$, we conclude:

$$\lambda_1^D(\Omega') \geq \lambda.$$

Specializing to $\Omega' = B_r$, the d -dimensional ball of radius r , recall the eigenvalue formula:

$$\lambda_1^D(B_r) = c_d r^{-2} \quad \text{where } c_d = j_{d/2-1,1}^2.$$

Domain monotonicity yields:

$$r \leq c_d \lambda^{-1/2}$$

This radius bound establishes the claimed $c_d\sqrt{\lambda}$ -density of the nodal set. $\square$

3.4. A lower bound on the size of the nodal set in dimension two. Nodal set can have isolated points by classical argument. Let us prove the following lower bound on the size of nodal sets for Dirichlet eigenfunctions of planar domains.

Theorem 3.4 (Brüning-Gromes [3]). *Let $\Omega \subset \mathbb{R}^2$ be a bounded domain, and let u_λ be an eigenfunction of $-\Delta_\Omega^D$ corresponding to an eigenvalue $\lambda > \lambda_1$. Then, the total length $L(\mathcal{Z}_{u_\lambda})$ of the nodal set satisfies*

$$L(\mathcal{Z}_{u_\lambda}) \geq C\sqrt{\lambda},$$

where $C > 0$ is a constant independent of λ .

Proof. Let c_2 be defined by equation $c_2 = \sqrt{\lambda_1(\mathbb{B}^2)}$, and partition Ω using a square grid with spacing

$$h := \frac{2c_2}{\sqrt{\lambda}}.$$

For each grid square $Q \subset \Omega$, consider the extended square $3Q$ (union of Q and its adjacent squares) with side length $3h$, as shown in Figure 5. Assume $3Q \subset \Omega$. By Theorem 3.3, there exists $p \in Q \cap \mathcal{Z}_{u_\lambda}$. If u_λ vanishes identically near p , the result is trivial (though impossible for analytic eigenfunctions). Otherwise, consider a nodal line through p :

Case 1: If the nodal line exits $3Q$, its length is at least h .

Case 2: If the nodal line remains in $3Q$, there is a nodal domain $\Omega' \subset 3Q$ with $p \in \partial\Omega'$. Let D_r be the minimal disk containing Ω' . By domain monotonicity :

$$\frac{c_2}{r^2} = \lambda_1^D(D_r) \leq \lambda_1^D(\Omega') = \lambda = \frac{4c_2^2}{h^2},$$

thus $r \geq h/2$. Therefore,

$$\text{Length}(\partial\Omega' \cap 3Q) \geq 2 \text{diam}(\Omega') \geq 2r \geq h.$$

In both cases, the nodal length in each $3Q$ is at least $h = 2c_2/\sqrt{\lambda}$. Since Ω contains $O(\lambda)$ such squares for large λ , there exists $C > 0$ with

$$L(\mathcal{Z}_{u_\lambda}) \geq C\sqrt{\lambda}. \quad \square$$

The high-dimensional case combines combinatorics and geometry and is very interesting. But it deserves a whole presentation because of its content.

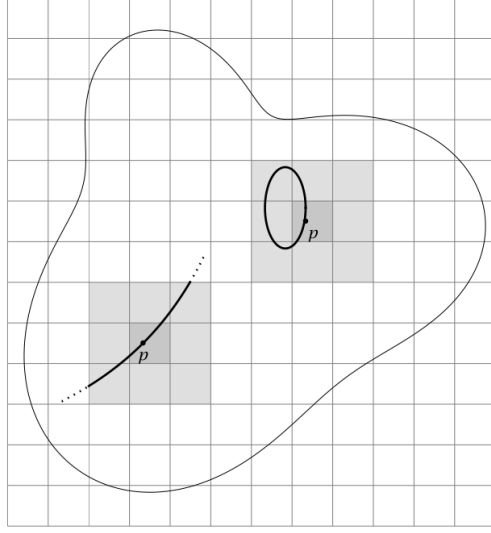


FIGURE 5. Grid structure: Original square Q (dark shading) and extended square $3Q$ (light shading). Left: Nodal line exits $3Q$. Right: Closed nodal line in $3Q$.

4. EIGENVALUE INEQUALITIES

4.1. Symmetric Decreasing Rearrangement. Given a set $A \subset \mathbb{R}^d$ of finite volume, its symmetric rearrangement is defined via the characteristic function $\chi_A^* := \chi_{A^*}$, where A^* denotes the ball centered at the origin with $\text{Vol}(A^*) = \text{Vol}(A)$. Let $u : \Omega \rightarrow \mathbb{R}$ be a non-negative measurable function on a bounded open set $\Omega \subset \mathbb{R}^d$. The symmetric decreasing

rearrangement of u is the function $u^* : \Omega^* \rightarrow \mathbb{R}$ defined by

$$u^*(x) = \int_0^{+\infty} \chi_{\gamma_u(t)}^*(x) dt,$$

where $\gamma_u(t) = \{x \in \Omega : u(x) > t\}$ are the superlevel sets of u .

This implies:

$$(4.1) \quad \|u\|_{L^p(\Omega)} = \|u^*\|_{L^p(\Omega^*)}.$$

4.2. Pólya–Szegő Principle. Let $\Omega \subset \mathbb{R}^d$ be a bounded domain, and let u be the first Dirichlet eigenfunction on Ω . Then:

$$(4.2) \quad \|\nabla u\|_{L^2(\Omega)} \geq \|\nabla u^*\|_{L^2(\Omega^*)}.$$

Proof :Pólya–Szegő inequality - Wikipedia

4.3. Faber–Krahn inequality.

Theorem 4.1 (Faber–Krahn inequality). *Let $\Omega \subset \mathbb{R}^d$ be a bounded domain, and Ω^* the ball of equal volume. Then*

$$\lambda_1^D(\Omega) \geq \lambda_1^D(\Omega^*),$$

where $\lambda_1^D(\cdot)$ denotes the first Dirichlet eigenvalue of the Laplacian.

Proof of Theorem 4.1. Utilizing equality (4.1) for the L^2 -norms and inequality (4.2), we establish that

$$R_\Omega[u] \geq R_{\Omega^*}[v],$$

where $u \in H_0^1(\Omega)$ represents the principal eigenfunction, and v denotes its symmetric decreasing rearrangement. We conclude that $v \in H_0^1(\Omega^*)$ via extending u by zero. Employing v as a test function in the characterization of $\lambda_1^D(\Omega^*)$, we obtain

$$\lambda_1^D(\Omega) = R_\Omega[u] \geq R_{\Omega^*}[v] \geq \lambda_1^D(\Omega^*),$$

which completes the proof. □

4.4. Dirichlet Cheeger’s Inequality. Let Ω be a compact Riemannian manifold with boundary or a bounded Euclidean domain. Then, the first Dirichlet eigenvalue of Ω , denoted $\lambda_1^D(\Omega)$, satisfies Cheeger’s inequality:

$$\lambda_1^D(\Omega) \geq \frac{1}{4} h_D^2(\Omega).$$

Proof. We have

$$(4.3) \quad \int_{\Omega} |\nabla \varphi| \, dx = \int_0^\infty \int_{L_\varphi(t)} d\Sigma_t \, dt = \int_0^\infty \text{Vol}_{d-1}(L_\varphi(t)) \, dt \geq h_D(\Omega) \int_0^\infty V_\varphi(t) \, dt = h_D(\Omega) \int_{\Omega} \varphi \, dx,$$

where the last equality follows from the layer-cake representation. Let u be the first Dirichlet eigenfunction of Ω . Then,

$$\int_{\Omega} |\nabla (u^2)| \, dx = 2 \int_{\Omega} |u| |\nabla u| \, dx \leq 2 \|u\|_{L^2(\Omega)} \|\nabla u\|_{L^2(\Omega)} = 2 \sqrt{\lambda_1^D(\Omega)} \|u\|_{L^2(\Omega)}^2.$$

Here, the first inequality is the Cauchy–Schwarz inequality, and the last equality holds since $\lambda_1^D(\Omega) = R[u]$, where $R[u]$ is the Rayleigh quotient of u .

We now apply (4.3) with $\varphi := u^2$, which yields:

$$\int_{\Omega} |\nabla (u^2)| \, dx \geq h_D(\Omega) \|u\|_{L^2(\Omega)}^2.$$

Combining this with the previous inequality, we obtain:

$$2 \sqrt{\lambda_1^D(\Omega)} \|u\|_{L^2(\Omega)}^2 \geq h_D(\Omega) \|u\|_{L^2(\Omega)}^2.$$

Since $\|u\|_{L^2(\Omega)}^2 > 0$, we can divide through by $\|u\|_{L^2(\Omega)}^2$, giving:

$$2 \sqrt{\lambda_1^D(\Omega)} \geq h_D(\Omega).$$

□

4.5. The Neumann Cheeger Constant. Let Ω be a compact Riemannian manifold (with or without boundary) or a bounded Euclidean domain of dimension d . The Neumann Cheeger constant is defined as

$$h_N := h_N(\Omega) = \inf_{\Gamma} \frac{\text{Vol}_{d-1}(\Gamma)}{\min\{\text{Vol}_d(\Omega_1), \text{Vol}_d(\Omega_2)\}},$$

where the infimum is taken over all smooth hypersurfaces Γ (not necessarily connected) that separate Ω into two open sets Ω_1 and Ω_2 .

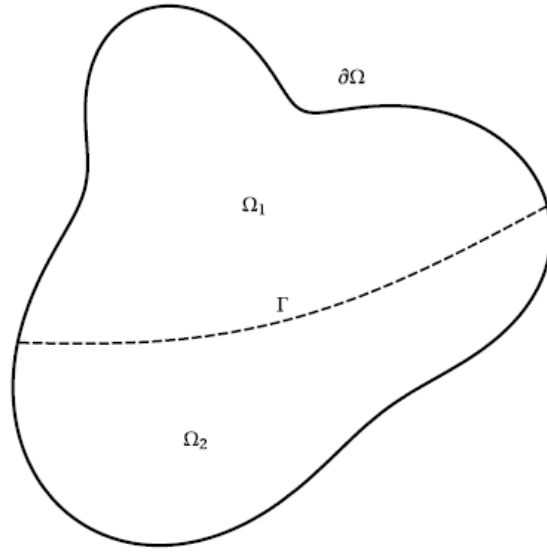


FIGURE 6.

4.6. Neumann Cheeger's Inequality. Let Ω be a compact Riemannian manifold with boundary or a bounded Lipschitz domain of dimension d , and let $\lambda_2^N(\Omega)$ be the first nonzero eigenvalue of the Neumann Laplacian $-\Delta_\Omega^N$. Then,

$$\lambda_2^N(\Omega) \geq \frac{1}{4} h_N^2(\Omega).$$

4.7. Cheeger's Inequality for Closed Manifolds. Let M be a closed Riemannian manifold of dimension d , and let $\lambda_1(M)$ be the first nonzero eigenvalue of the Laplace–Beltrami operator $-\Delta_M$. Then,

$$\lambda_1(M) \geq \frac{1}{4} h_N^2(M),$$

where $h_N(M)$ is the Neumann Cheeger constant.

4.8. Hyperbolic Space Example. We begin with $\mathbb{H}^d$, the hyperbolic space of constant sectional curvature -1 . Consider a geodesic ball $B_r \subset \mathbb{H}^d$ of radius r . We're told that an explicit computation shows:

$$\frac{\text{Vol}_{d-1}(\partial B_r)}{\text{Vol}_d(B_r)} > d - 1$$

for any $r > 0$. Here, $\text{Vol}_{d-1}(\partial B_r)$ is the $(d-1)$ -dimensional volume of the boundary of the ball (a sphere of radius r), and $\text{Vol}_d(B_r)$ is the d -dimensional volume of the ball itself. Thus, from the given computation:

$$h_D(B_r) > d - 1$$

for all $r > 0$. Now, consider the first Dirichlet eigenvalue of B_r , denoted $\lambda_1^D(B_r)$, which corresponds to the smallest eigenvalue of the Laplacian with Dirichlet boundary conditions (i.e., functions vanish on ∂B_r). We're given:

$$\lambda_1^D(B_r) = \frac{(d-1)^2}{4} + O\left(\frac{1}{r^2}\right) \quad \text{as } r \rightarrow \infty.$$

This asymptotic behavior indicates that as the radius grows large, the eigenvalue approaches $\frac{(d-1)^2}{4}$, with a correction term that decays like $\frac{1}{r^2}$.

5. SPECTRAL INVARIANTS

5.1. Heat Kernel on a Riemannian Manifold. Let (M, g) be a closed Riemannian manifold. Consider the initial-value problem for the heat equation:

$$\begin{cases} \frac{\partial u}{\partial t}(t, y) = \Delta_y u(t, y), & t \in \mathbb{R}_+ = (0, +\infty), y \in M, \\ u(0, y) = \varphi(y), & y \in M, \end{cases}$$

where $\varphi(y)$ represents an initial temperature distribution or diffusing substance density.

Definition 5.1. A fundamental solution (heat kernel) is a function $e(t, x, y) \in C(\mathbb{R}_+ \times M \times M)$, C^1 in t and C^2 in y , satisfying heat equation with $\varphi(y) = \delta_x(y)$. The initial condition holds weakly:

$$\lim_{t \rightarrow 0^+} \int_M e(t, x, y) f(y) dy = f(x), \quad \forall f \in C(M).$$

Theorem 5.1. For a closed (M, g) , there exists a unique smooth heat kernel:

$$(6.1.3) \quad e(t, x, y) = \sum_{j=0}^{\infty} e^{-\lambda_j t} u_j(x) u_j(y),$$

where $\{u_j\}$ are orthonormal eigenfunctions of $-\Delta_M$ with eigenvalues λ_j .

5.2. Isospectrality and Spectral Rigidity. Two closed Riemannian manifolds (M, g) and (N, h) are called **isospectral** if their Laplace–Beltrami operators satisfy

$$\text{Spec}(-\Delta_{(M,g)}) = \text{Spec}(-\Delta_{(N,h)}),$$

where the spectrum is understood as a multiset with multiplicities preserved. This definition extends to manifolds with boundary or Euclidean domains by specifying boundary conditions (e.g., Dirichlet, Neumann).

The heat trace $e_M(t) = \sum_{j=0}^{\infty} e^{-\lambda_j t} = \int e(t, x, x) dx$ serves as a critical tool in studying isospectrality. A key observation is: Isospectral manifolds share identical heat traces:

$$e_M(t) = e_N(t) \quad \forall t > 0.$$

5.3. Poisson Summation Formula. For a Schwartz function $f \in \mathcal{S}(\mathbb{R}^d)$,

$$\sum_{k \in \mathbb{Z}^d} f(k) = \sum_{m \in \mathbb{Z}^d} \widehat{f}(m),$$

where the Fourier transform is defined as

$$\widehat{f}(y) = (2\pi)^{-d/2} \int_{\mathbb{R}^d} f(x) e^{-i\langle x, y \rangle} dx.$$

For a lattice $\Gamma \subset \mathbb{R}^d$, let Γ^* denote its dual. The generalized Poisson formula states:

$$\sum_{k \in \Gamma} f(k) = \frac{1}{\text{Vol}(\Gamma)} \sum_{m \in \Gamma^*} \widehat{f}(m),$$

where $\text{Vol}(\Gamma) = \text{Vol}(\mathbb{R}^d/\Gamma)$. For $f(x) = e^{-a|x|^2}$, $a > 0$,

$$\widehat{f}(y) = \left(\frac{\pi}{a}\right)^{d/2} e^{-\frac{\pi^2|y|^2}{a}}.$$

Substituting $a = \frac{1}{4t}$ yields:

$$\sum_{x \in \Gamma^*} e^{-4\pi^2 t |x|^2} = \frac{\text{Vol}(\Gamma)}{(4\pi t)^{d/2}} \sum_{y \in \Gamma} e^{-\frac{|y|^2}{4t}}.$$

Left-hand side: Heat trace of the flat torus $\mathbb{R}^d/\Gamma$, with eigenvalues $4\pi^2|x|^2$, $x \in \Gamma^*$. Right-hand side: Sum over closed geodesics in $\mathbb{R}^d/\Gamma$. This links spectral (quantum) and dynamical (classical) quantities, a manifestation of Bohr's correspondence principle.

5.4. Special Lattices. For $d = 8k$, $k \in \mathbb{N}$:

- $\Gamma_2 \subset \mathbb{Z}^d$: Sublattice of index 2, where $\sum_{j=1}^d x_j$ is even.
- $\Gamma^{(d)}$: Generated by Γ_2 and $w_d = (\frac{1}{2}, \dots, \frac{1}{2})$.
- $\Gamma_2 \subset \Gamma^{(d)}$ has index 2, $\text{Vol}(\Gamma^{(d)}) = 1$.

For $\Gamma = \Gamma^{(d)}$ ($d = 8k$):

- $|x|^2$ is even for all $x \in \Gamma$.
- $\Gamma^* = \Gamma$.

5.5. Milnor's construction.

Theorem 5.2. *The 16-dimensional tori $M_1 = \mathbb{R}^{16}/\Gamma^{(16)}$ and $M_2 = \mathbb{R}^{16}/\Gamma^{(8,8)}$ ($\Gamma^{(8,8)} = \Gamma^{(8)} \oplus \Gamma^{(8)}$) are isospectral but non-isometric.*

Non-isometry: $\Gamma^{(8)}$ contains vectors of norm $\sqrt{2}$, while $\Gamma^{(16)}$ does not.

Isospectrality: Compare heat traces via theta functions. For $\Gamma \subset \mathbb{Z}^{16}$:

$$\theta_\Gamma(t) := e_{\mathbb{R}^d/\Gamma} \left(\frac{t}{4\pi} \right) = \sum_{x \in \Gamma^*} e^{-\pi|x|^2 t},$$

where Γ^* is the dual lattice. For $\Gamma \subset \mathbb{Z}^{16}$ satisfying:

- (1) $|x|^2 \in 2\mathbb{Z}$ for all $x \in \Gamma$.
- (2) $\Gamma^* = \Gamma$.

the Poisson summation formula yields:

$$\theta_\Gamma(t) = t^{-8} \sum_{y \in \Gamma} e^{-\pi|y|^2/t}.$$

This implies the functional equation:

$$\theta_\Gamma(t) = t^{-8} \theta_\Gamma(t^{-1}).$$

5.6. Holomorphic Extension and Modular Forms. Analytic Continuation: By the Weierstrass identity theorem, $\theta_\Gamma(z)$ extends holomorphically to $\text{Re}(z) > 0$ and satisfies:

$$\theta_\Gamma(z) - z^{-8}\theta_\Gamma(z^{-1}) = 0 \quad \forall z \in \mathbb{C}, \text{Re}(z) > 0.$$

Modular Transformation: Define $\tilde{\theta}_\Gamma(z) = \theta_\Gamma(-iz)$ for $z = u+iv$ ($v > 0$). This function:

- Is holomorphic in the upper half-plane $\mathbb{H} = \{z : \text{Im}(z) > 0\}$,
- Satisfies the modularity condition:

$$\tilde{\theta}_\Gamma(z+1) = \tilde{\theta}_\Gamma(z),$$

- Decays asymptotically: $\tilde{\theta}_\Gamma(u+iv) \rightarrow 1$ as $v \rightarrow \infty$.

The space of weight-8 modular forms is one-dimensional. Since $\tilde{\theta}_\Gamma(z)$ is uniquely determined and its asymptotic behavior must coincide for both $\Gamma^{(16)}$ and $\Gamma^{(8,8)}$. Thus:

$$\theta_{\Gamma^{(16)}}(t) = \theta_{\Gamma^{(8,8)}}(t) \quad \forall t > 0.$$

5.7. Sunada's Construction. Worth a single presentation.

5.8. Spectral Invariants. From the viewpoint of spectral geometry, particular interest lies in the behavior of the heat kernel on the diagonal $x = y$ as $t \rightarrow 0^+$.

Theorem 5.3 (Minakshisundaram–Pleijel Asymptotic Expansion). *Let (M, g) be a closed Riemannian manifold with $\dim M = d$. The heat kernel admits the asymptotic expansion:*

$$e(t, x, x) = (4\pi t)^{-\frac{d}{2}} \left(\sum_{j=0}^k a_j(x) t^j + O(t^{k+1}) \right),$$

for all $k > 0$. The coefficients $a_j(x)$, called **local heat invariants**, are determined by the local geometry of M near x .

Using the spectral decomposition $e(t, x, x) = \sum_{j=0}^{\infty} e^{-\lambda_j t} u_j(x)^2$, the short-time asymptotics imply:

$$(6.1.12) \quad \sum_{j=0}^{\infty} e^{-\lambda_j t} = (4\pi t)^{-\frac{d}{2}} \left(\sum_{j=0}^{\infty} a_j t^j \right), \quad t \rightarrow 0^+,$$

where $a_j = \int_M a_j(x) dx$. These global heat invariants a_j encode geometric information about M .

Explicit calculations in Riemannian normal coordinates yield:

- $a_0(x) = 1$, hence $a_0 = \text{Vol}(M)$
- $a_1(x) = \frac{1}{6}\tau(x)$, where $\tau(x)$ is the scalar curvature and $a_1 = \frac{1}{6} \int_M \tau(x) dx$

For surfaces ($d = 2$), the Euler characteristic is spectrally determined via:

$$\chi(M) = \frac{1}{4\pi} \int_M \tau(x) dx = \frac{3}{2\pi} a_1(M).$$

Thus, one can "hear" the topology of a drum—the number of handles on an orientable surface is a spectral invariant.

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